

Variables

Problem 1: Variables of States

Suppose that the fifty states are the observational units. For each item, write **yes, it is a variable** or **no, it is not a variable**.

For each variable, report or make a reasonable guess for its value in your home state. For each item that is not a variable, briefly explain why.

- a) The number of states that have a female governor
- b) The percentage of the state's residents who are over 65 years of age
- c) The highest speed limit in the state
- d) Whether the state's name consists of one word
- e) The average income of an adult resident of the state
- f) The number of births per 1,000 residents of the state in 1999
- g) The number of states that were settled before 1865
- h) The average number of Congressional representatives per state
- i) The number of Congressional representatives from the state

Problem 2: Types of Variables

The observational units are 22 Broadway productions. Classify each variable from the supplied dataset as **quantitative** or **categorical**. If a categorical variable has exactly two possible values, also identify it as **binary**.

- a) Production
- b) Type
- c) Gross
- d) Attendance
- e) Capacity
- f) Top price